

Energy-Efficient Logic-in-Memory

Enabled by a Physics-Based RRAM Compact Model

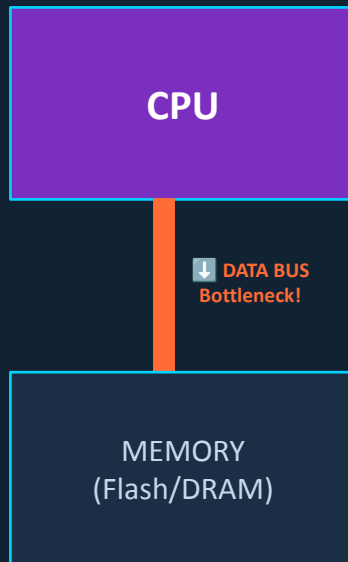
PRESENTER

Jainil Devaliya (23110090)
Madhup Sankhla (23110190)
Adarsh Patidar (23110012)

Motivation — The Computing Bottleneck

PROBLEM STATEMENT

- ▶ **Von Neumann bottleneck limits compute speed**
- ▶ Memory & CPU are physically separated
- ▶ Constant data transfer → high energy waste
- ▶ IoT devices need ultra-low power operation
- ▶ **Flash memory access: $\sim 50 \mu\text{J}$ / $\sim 0.5 \text{ ms}$ per op**



$\sim 10^9 \times$ energy gap: compute vs memory access

Why RRAM? — The Ideal Device

DEVICE MOTIVATION

Non-Volatile

Retains state without power
Ideal for memory + logic

Ultra-Fast

Switching < 300 ps
Nitride-based < 100 ps

Dual Function

Acts as memory AND logic
Eliminates data movement

IoT Ready

Low power, compact footprint
 $\leq 4F^2$ cell area

Physics-Based RRAM Compact Model

DEVICE MODEL

- ▶ Implemented in Verilog-A
- ▶ Reproduces AC + DC behavior accurately
- ▶ Valid for ultra-short pulses (10 ns)
- ▶ Includes intrinsic resistance variability (Gaussian)
- ▶ Models temperature & conductive filament dynamics

HRS (Logic 0)

Broken
filament

R: 30–70 k Ω

VRESET = -1.3 V

LRS (Logic 1)

Intact
filament (CF)

R: 3–7 k Ω

IC = 100 μ A

(Fig. 1c – RRAM device schematic)

KEY MODEL EQUATIONS

$$R = R_{LRS} \cdot [(\tau_{ox} - x)/\tau_{ox} + \exp(x/k - 1) \cdot e^{(EG/kBT)}]$$

$$I = (V_0/R) \cdot \sinh(V/V_0)$$

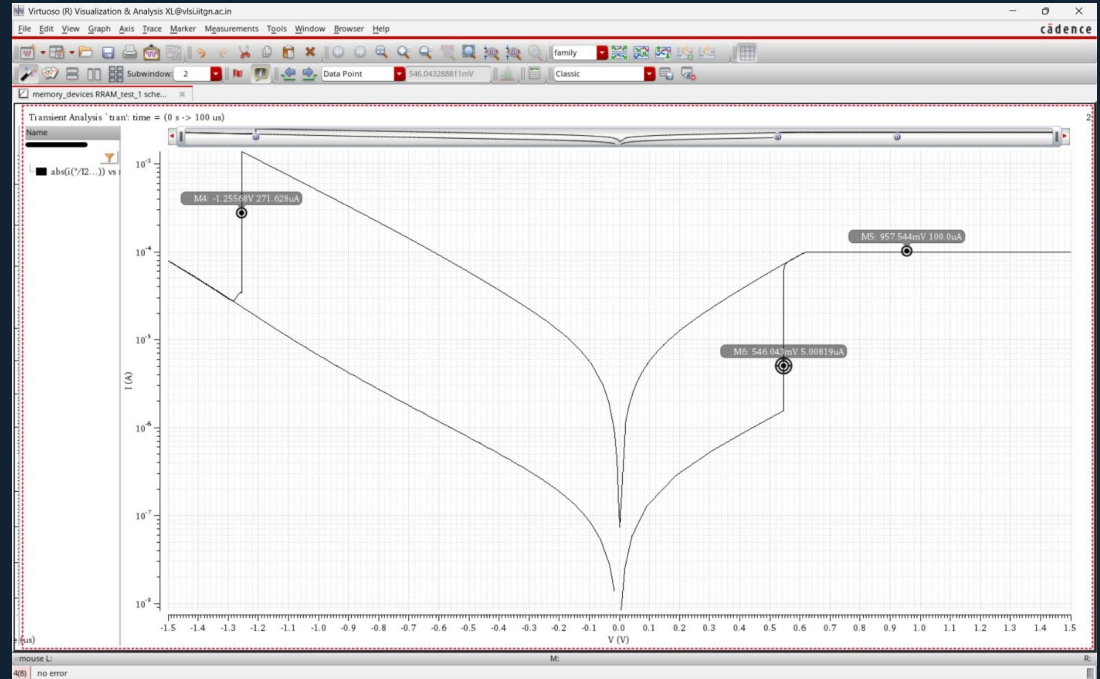
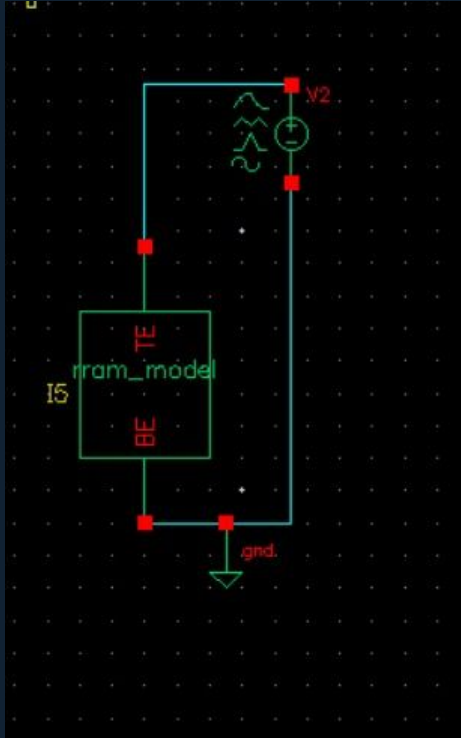
$$dx/dt = c_0 \cdot \exp[(ED - (g_0 - \alpha x)^\beta \cdot T_{ox}) / kBT] \quad (\text{reset})$$

$$dx/dt = -xc_0 \cdot \exp[(EG - f \cdot V/x) / kBT] \quad (\text{set})$$

$$dT/dt = Cp^{-1} \cdot [V \cdot I - kT(T - T_0)]$$

RRAM

DEVICE MODEL

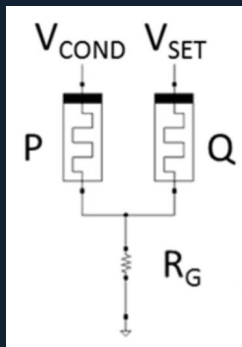


Paper Reset: -1.3, Set: 0.6
Our Reset: -1.255, Set: 0.546

IMPLY Logic — The Building Block

LOGIC SCHEME

- ▶ **Material Implication: $P \text{ IMPLY } Q = (\neg P \vee Q)$**
- ▶ IMPLY + FALSE = complete Boolean logic set
- ▶ Two RRAM devices required per gate
- ▶ FALSE = unconditional RESET to logic 0
- ▶ All combinational circuits constructible from IMPLY



VSET = 1.16 V | VCOND = 1.05 V | RG = 4 kΩ

IMPLY TRUTH TABLE

P	Q	Q' (Output)
0	0	1
0	1	1
1	0	0
1	1	1

IMPLY Operation — How It Works

OPERATION DETAIL



OPERATION TIMELINE (10 ns pulse)

P (RP):



Stays at 10 kΩ (Logic 0)

Q (RQ):



RESET → SET
(Logic 0 → 1)

Logic 1 band: 3 – 7 kΩ

Logic 0 band: 30 – 70 kΩ

Logic State Degradation — Key Challenge

RELIABILITY

- ▶ **Repeated IMPLY ops cause resistance drift**
- ▶ P=1, Q=0: Q experiences unwanted set voltage
- ▶ Progressive RQ decrease → eventual bit flip
- ▶ **Non-optimized: error after only ~67 cycles**
- ▶ **Optimized voltages: stable for >3000 cycles**

Initial RQ = 60 k Ω (worst-case variability scenario)

VOLTAGE OPTIMIZATION IMPACT

✗ Non-Optimized

VSET = 1.12 V, VCOND = 0.88 V
→ Fails after 67 IMPLY cycles

✓ Optimized

VSET = 1.16 V, VCOND = 1.05 V
→ Stable for >3000 IMPLY cycles

(RQ degradation over repeated operations)

1-Bit Full Adder Design

IMPLEMENTATION

- ▶ **9 RRAM devices in crossbar array**
- ▶ Inputs: A, B, Cin (logic-state preserved)
- ▶ Outputs: S (sum) in M1, Cout (carry) in M5
- ▶ Auxiliary: M2, M3, M4, M6 (temporary states)
- ▶ **43 total steps: 17 FALSE + 26 IMPLY ops**

9-DEVICE LAYOUT



(Fig. 5a–b – Full adder schematic and truth table)

Proposed Innovations & Future Directions

01 Reduce FALSE Operations

New logic sequences minimizing reset steps
Estimated 2–3× energy reduction possible

02 Adaptive Voltage Control

Dynamic VSET/VCOND tuning per cycle
Counters logic state degradation actively

03 Hybrid CMOS + RRAM

CMOS for control, RRAM for data ops
Best-of-both-worlds architecture

04 Parallel IMPLY Execution

Simultaneous multi-gate computation
Reduces 43-step overhead significantly

05 Error Correction Layer

ECC to mask variability-induced flips
Extends reliable operation cycles

06 3D Crossbar Stacking

Vertical integration for higher density
Sub-1 fJ/op target with 3D RRAM